

Genome-wide genetic correlation across sleep and frailty

Primary Frailty Index plus seven latent dimensions; estimates are grouped by their frozen correction family

A Sleep × frailty rg estimates

Cell value = rg; * indicates $q \leq 0.05$ within that column's own family
FI q-values: frozen all-396 atlas family | Latent-factor q-values: fixed 84-pair sensitivity family



No direct tests compare estimates between FI and latent factors; differences in color/significance are descriptive.

B Robustness and replication

PRIMARY FI

9/12

pass inherited all-396 BH $q \leq 0.05$; 9/12
also pass same-family Bonferroni
sensitivity

LATENT FACTORS

48/84

pass BH $q \leq 0.05$ in the fixed secondary
family; sensitivity-only, exact overlap
unknown

INDEPENDENT PAIRWISE REPLICATION

None established

Audited candidates do not provide an
eligible independent sleep–frailty rg
pair

COHORT OVERLAP

Unresolved

UK Biobank overlap is expected in 11/12
primary pairs and in latent sources;
exact intersections unknown

LDSC rg is genome-wide covariance,
not causation or a causal mechanism.

INTERPRETATION

The FI values reuse frozen atlas results; the seven-factor family is secondary and sensitivity-only.
The q-value denominators differ by column. No plotted pattern establishes independent replication, local sharing, or which frailty dimension drives overlap.
See the source-data TSV for estimates, SEs, P values, family labels, overlap status and interpretation limits.